

Workforce Data Analysis: Salary and Remote Work Trends Across Job Titles and Company Locations

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Tool Used: SQL Server

Domain: Workforce Analytics

Introduction

- Organizations need workforce analytics to make informed hiring and compensation decisions.
- Salary trends and remote work adoption have become critical business factors.
- This project analyzes workforce salary data across multiple countries, experience levels, and job roles.
- SQL Server was used to perform business-focused analysis.

Problem Statement

Objectives

- Analyze salary trends across job titles.
- Study remote work adoption.
- Compare salaries by experience level.
- Identify high-paying countries and roles.
- Support workforce planning and talent retention.
- Generate business insights through SQL analysis.

Dataset Description

Dataset Attributes

- work_year
- experience_level
- employment_type
- job_title
- salary
- salary_in_usd
- employee_residence
- remote_ratio
- company_location
- company_size

Experience Levels

- EN – Entry Level
- MI – Mid Level
- SE – Senior Level
- EX – Executive Level

Methodology

Project Workflow

1. Data Collection
2. Data Understanding
3. Data Cleaning
4. SQL Query Execution
5. Business Analysis
6. Insight Generation
7. Recommendation Development

SQL Concepts Used

- SELECT
- WHERE
- GROUP BY
- HAVING
- CASE
- UPDATE
- Aggregate Functions
- Subqueries

Salary Analysis

Analysis Performed

1. Employees by Company Size in 2021

```
SELECT company_size,  
       COUNT(*) AS employee_count  
FROM workforce_data  
WHERE work_year = 2021  
GROUP BY company_size;
```

2. Top 3 Highest Paying Part-Time Jobs in 2023

```
SELECT TOP 3 job_title,  
       AVG(salary_in_usd) AS avg_salary  
FROM workforce_data  
WHERE employment_type='PT'  
AND work_year=2023  
GROUP BY job_title  
HAVING COUNT(*) > 50  
ORDER BY avg_salary DESC;
```

3. Countries Where Mid-Level Salary is Above Global Mid-Level Average

```
SELECT employee_residence,  
       AVG(salary_in_usd) AS avg_salary  
FROM workforce_data  
WHERE experience_level='MI'  
AND work_year=2023  
GROUP BY employee_residence  
HAVING AVG(salary_in_usd) >  
(  
  SELECT AVG(salary_in_usd)  
  FROM workforce_data  
  WHERE experience_level='MI'  
  AND work_year=2023  
);
```

4. Highest and Lowest Paying Countries for Senior Employees

```
SELECT TOP 1 company_location,  
       AVG(salary_in_usd) avg_salary  
FROM workforce_data  
WHERE experience_level='SE'  
AND work_year=2023  
GROUP BY company_location  
ORDER BY avg_salary DESC;
```

5. Salary Growth by Job Title

```
SELECT job_title,  
AVG(CASE WHEN work_year=2023 THEN salary_in_usd END) AS salary_2023,  
AVG(CASE WHEN work_year=2024 THEN salary_in_usd END) AS salary_2024  
FROM workforce_data  
GROUP BY job_title;
```

6. Top 3 Countries with Highest Entry-Level Growth

```
SELECT TOP 3 employee_residence,  
(  
AVG(CASE WHEN work_year=2023 THEN salary_in_usd END)  
-  
AVG(CASE WHEN work_year=2020 THEN salary_in_usd END)  
)  
/  
AVG(CASE WHEN work_year=2020 THEN salary_in_usd END)  
*100 AS growth_percent  
FROM workforce_data  
WHERE experience_level='EN'  
GROUP BY employee_residence  
HAVING COUNT(*)>50  
ORDER BY growth_percent DESC;
```

7. Update Remote Ratio

```
UPDATE workforce_data  
SET remote_ratio = 100  
WHERE salary_in_usd > 90000  
AND employee_residence IN ('US','AU');
```

8. Salary Update by Experience Level

```
UPDATE workforce_data  
SET salary_in_usd =  
CASE  
WHEN experience_level='SE'  
THEN salary_in_usd*1.22  
  
WHEN experience_level='MI'  
THEN salary_in_usd*1.30  
  
WHEN experience_level='EN'  
THEN salary_in_usd*1.18  
  
WHEN experience_level='EX'  
THEN salary_in_usd*1.25  
  
ELSE salary_in_usd  
END;
```


9. Year with Highest Salary for Each Job

```
SELECT job_title,  
work_year,  
AVG(salary_in_usd) avg_salary  
FROM workforce_data  
GROUP BY job_title,work_year  
ORDER BY job_title,avg_salary DESC;
```

10. Percentage of Employment Types

```
SELECT job_title,  
  
100.0*  
SUM(CASE WHEN employment_type='FT' THEN 1 END)  
/COUNT(*) AS FT_Percentage,  
  
100.0*  
SUM(CASE WHEN employment_type='PT' THEN 1 END)  
/COUNT(*) AS PT_Percentage  
  
FROM workforce_data  
GROUP BY job_title;
```

11. Remote Managers Above \$90K

```
SELECT employee_residence,  
COUNT(*) total_managers  
FROM workforce_data  
WHERE job_title LIKE '%Manager%'  
AND salary_in_usd > 90000  
AND remote_ratio=100  
GROUP BY employee_residence;
```

12. Top 5 Countries with Large Companies

```
SELECT TOP 5 company_location,  
COUNT(*) total_large_companies  
FROM workforce_data  
WHERE company_size='L'  
GROUP BY company_location  
ORDER BY total_large_companies DESC;
```

13. Percentage of Fully Remote Employees Above \$100K

```
SELECT  
(  
COUNT(*)*100.0  
/  
(SELECT COUNT(*) FROM workforce_data)  
) AS percentage  
FROM workforce_data  
WHERE remote_ratio=100  
AND salary_in_usd>100000;
```


14. Locations with Entry-Level Salary Above Market Average

```
SELECT employee_residence,  
AVG(salary_in_usd) avg_salary  
FROM workforce_data  
WHERE experience_level='EN'  
GROUP BY employee_residence  
HAVING AVG(salary_in_usd)  
>  
(  
SELECT AVG(salary_in_usd)  
FROM workforce_data  
WHERE experience_level='EN'  
);
```

15. Highest Paying Country for Each Job Title

```
SELECT job_title,  
employee_residence,  
MAX(salary_in_usd) highest_salary  
FROM workforce_data  
GROUP BY job_title,employee_residence;
```

16. Countries with Consistent Salary Growth

```
SELECT employee_residence  
FROM workforce_data  
GROUP BY employee_residence  
HAVING  
AVG(CASE WHEN work_year=2021 THEN salary_in_usd END)  
<  
AVG(CASE WHEN work_year=2022 THEN salary_in_usd END)  
  
AND  
  
AVG(CASE WHEN work_year=2022 THEN salary_in_usd END)  
<  
AVG(CASE WHEN work_year=2023 THEN salary_in_usd END);
```

17. Remote Work Comparison (2021 vs 2024)

```
SELECT work_year,  
experience_level,  
  
100.0*  
SUM(CASE WHEN remote_ratio=100 THEN 1 ELSE 0 END)  
/  
COUNT(*) AS remote_percentage  
  
FROM workforce_data  
  
WHERE work_year IN (2021,2024)  
  
GROUP BY work_year,experience_level;
```

18. Salary Increase by Experience Level & Job Title

```
SELECT experience_level,  
job_title,  
  
AVG(CASE WHEN work_year=2024 THEN salary_in_usd END)  
-  
AVG(CASE WHEN work_year=2023 THEN salary_in_usd END)  
  
AS increase_amount  
  
FROM workforce_data  
  
GROUP BY experience_level,job_title;
```

19. Role-Based Access Control

Finding:

Access should be restricted based on experience level.

Example:

- Entry Level → View only Entry-Level records
- Mid Level → View MI records
- Senior Level → View SE records
- Executive → View all records

20. Domain Switching Recommendations

Recommendations:

Employees can transition toward:

- AI Engineer
- Machine Learning Engineer
- Data Scientist
- Data Engineer
- MLOps Engineer

These roles generally show stronger salary growth and demand trends according to workforce analytics.

Key Findings :

Major Findings

- Senior professionals earn the highest salaries.
- Remote work opportunities continue to grow.
- Large companies dominate workforce hiring.
- Salary levels vary significantly across countries.
- Data-focused roles remain highly rewarding.

Business Insights & Recommendations

Recommendations

- Implement competitive salary benchmarking.
- Expand recruitment in high-growth regions.
- Increase remote work opportunities.
- Focus on employee retention programs.
- Invest in upskilling for emerging technology roles.

Business Value

- Better workforce planning
- Improved talent acquisition
- Enhanced employee retention
- Data-driven HR decisions

Conclusion

The Workforce Data Analysis project successfully demonstrated how SQL can be used to analyze salaries, remote work trends, and workforce distribution.

The analysis provided valuable insights into compensation structures, hiring trends, and remote work adoption, helping organizations make informed strategic decisions and improve workforce management.

Thank You